

BERGERAC DORDOGNE PERIGORD AP, France

WMO: 075300

Lat: 44.8231N

Lon: 0.5242E

Elev: 171

StdP: 14.61

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 22.5	(c) 25.5	(d) 17.8	(e) 13.6	(f) 27.1	(g) 21.3	(h) 16.1	(i) 29.6	(j) 21.9	(k) 49.8	(l) 19.1	(m) 50.7	(n) 2.8	(o) 70	(p) 0.461

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 23.1	(c) 91.9	(d) 71.3	(e) 88.2	(f) 69.7	(g) 85.1	(h) 68.5	(i) 73.4	(j) 87.2	(k) 71.8	(l) 85.2	(m) 70.1	(n) 82.1	(o) 6.1	(p) 150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a) 69.1	(b) 107.6	(c) 79.7	(d) 67.3	(e) 101.0	(f) 77.5	(g) 65.8	(h) 95.7	(i) 75.5	(j) 37.0	(k) 87.1	(l) 35.5	(m) 85.4	(n) 34.1	(o) 82.0	(p) 81.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 18.1	(b) 15.5	(c) 13.4	(e) 17.1	(f) 98.0	(g) 4.6	(h) 3.2	(i) 13.7	(j) 100.3	(k) 11.0	(l) 102.2	(m) 8.4	(n) 104.0	(o) 5.1	(p) 106.3		
			DB	WB	DB	WB	DB	WB	DB	WB	DB	WB	DB	WB		
			17.1	98.0	4.6	3.2	13.7	100.3	11.0	102.2	8.4	104.0	5.1	106.3		
			16.5	76.5	4.5	1.8	13.2	77.8	10.6	78.9	8.0	79.9	4.8	81.3		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	55.4	42.4	43.3	48.9	53.3	59.7	66.6	69.9	69.1	63.1	57.5	47.8	42.8		
	DBStd	11.68	6.81	6.74	5.73	5.40	5.81	6.08	5.13	5.56	5.57	6.72	6.64	7.28		
	HDD50	932	248	199	87	26	3	0	0	0	0	17	116	236		
	HDD65	3975	701	608	498	351	182	53	14	21	101	244	515	687		
	CDD50	2921	11	11	55	126	304	497	617	593	392	250	51	14		
	CDD65	488	0	0	0	1	18	100	166	149	43	11	0	0		
	CDH74	5880	0	0	5	61	340	1207	1870	1684	616	97	0	0		
	CDH80	2151	0	0	0	6	73	440	746	696	178	12	0	0		
(14) Wind	WSAvg	5.6	5.8	6.3	6.6	6.2	5.7	5.5	5.4	4.9	4.6	4.9	5.5	5.5		
(15) Precipitation	PrecAvg	32.3	2.9	2.3	2.2	3.2	3.0	2.6	2.2	2.4	2.6	2.8	3.2	2.9		
	PrecMax	39.7	7.5	5.4	5.9	6.5	6.0	5.7	5.6	6.8	6.6	6.9	8.6	7.8		
	PrecMin	21.4	0.7	0.2	0.4	0.5	0.4	0.4	0.5	0.5	0.9	0.9	1.1	0.5		
	PrecStd	4.9	1.7	1.5	1.3	1.8	1.3	1.3	1.2	1.5	1.5	1.4	1.7	1.7		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	60.2	65.5	74.0	80.6	86.0	94.6	96.2	98.7	89.5	81.1	69.3	61.7		
		MCWB	54.0	54.1	56.5	62.5	68.0	72.9	71.6	73.1	68.5	66.6	59.8	54.9		
	2%	DB	57.2	60.7	68.8	75.1	82.0	88.6	91.0	92.5	84.9	76.5	64.5	58.3		
		MCWB	52.9	52.1	55.6	59.9	65.5	69.9	71.3	71.0	67.0	64.7	57.9	53.6		
	5%	DB	54.9	57.2	64.9	71.1	78.0	85.0	87.7	87.6	81.1	72.7	61.4	55.9		
		MCWB	51.5	50.6	53.7	58.0	63.9	68.6	70.0	69.7	65.8	63.1	56.4	52.6		
	10%	DB	52.7	54.2	61.2	66.6	73.7	81.2	84.2	83.3	77.0	69.3	58.5	53.9		
		MCWB	49.8	49.8	52.0	55.8	61.7	67.0	68.6	68.2	64.3	61.6	54.9	51.1		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	55.8	55.7	59.1	64.5	70.4	74.9	74.9	75.6	72.3	68.5	61.3	56.5		
		MCDB	58.1	62.4	69.5	77.0	82.2	88.3	88.6	89.8	84.2	77.2	66.5	58.8		
	2%	WB	53.8	53.5	56.8	61.6	67.1	72.2	73.1	73.2	69.3	66.1	59.1	54.6		
		MCDB	56.5	58.0	66.1	73.1	78.9	86.1	86.8	87.1	80.5	73.5	62.8	57.3		
	5%	WB	51.9	51.9	55.0	58.9	64.8	70.0	71.5	71.4	67.2	64.2	57.0	53.0		
		MCDB	54.3	55.8	62.4	68.4	75.1	81.9	85.0	84.7	77.3	70.7	60.5	55.5		
	10%	WB	50.1	50.3	53.2	56.7	62.7	67.8	69.8	69.4	65.3	62.4	55.1	51.3		
		MCDB	52.4	53.8	59.6	65.0	71.9	78.5	81.8	80.8	74.4	68.3	58.4	53.8		
(35) Mean Daily Temperature Range	5% DB	MDBR	14.1	18.0	21.4	21.8	21.9	22.4	23.1	24.3	24.6	20.8	16.1	14.8		
		MCDBR	16.3	24.0	30.6	31.4	29.9	30.0	31.1	33.7	32.0	25.1	19.8	16.9		
		MCWBR	12.2	16.0	18.1	17.5	15.3	13.7	13.4	14.3	15.5	14.3	13.7	12.5		
	5% WB	MCDBR	13.1	19.3	24.8	28.0	26.2	26.9	27.3	28.4	26.8	21.3	17.1	14.7		
		MCWBR	10.5	13.7	15.3	16.1	14.6	12.9	12.5	13.1	13.9	12.7	12.5	11.5		
(40) Clear-Sky Solar Irradiance	taub		0.323	0.337	0.371	0.401	0.405	0.419	0.409	0.406	0.375	0.367	0.340	0.321		
	taud		2.481	2.422	2.333	2.248	2.291	2.289	2.313	2.335	2.402	2.437	2.461	2.479		
	Ebn at Noon		252	268	272	272	275	271	272	268	268	253	241	239		
	Edh at Noon		22	28	35	41	41	41	40	37	32	27	23	20		
(44) All-Sky Solar Radiation	RadAvg		415	716	1085	1445	1686	1938	1909	1692	1356	864	495	380		
	RadStd		48	83	126	172	137	164	141	116	86	71	59	54		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (25 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page